

ECSE-4040

Digital Electronics

Lecture 3

Prof. T. Paul Chow

Bldg. CII, Room 6111

Rensselaer Polytechnic Institute

Troy, NY 12180-3590

Tel. (518)276-2910

FAX (518)276-2990

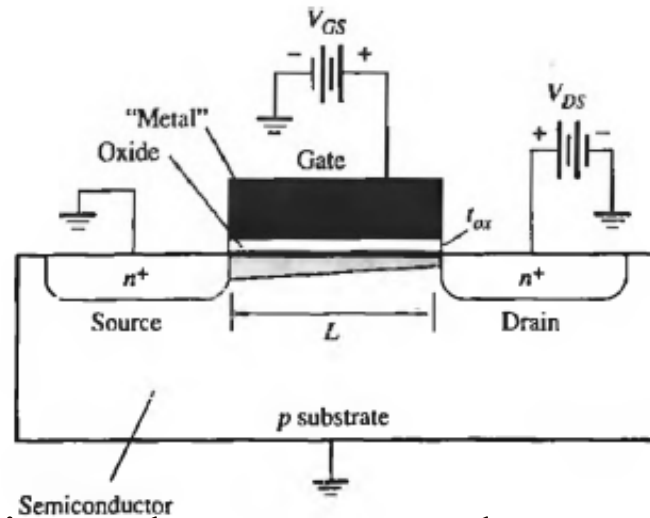
e-mail: chowt@rpi.edu

<https://lms.rpi.edu>

Lecture Outline

- MOSFET Structure and Operation
- Flatband and Threshold Voltages
- I-V Characteristics
 - Long Channel
 - Short Channel
- Gate and Junction Capacitances

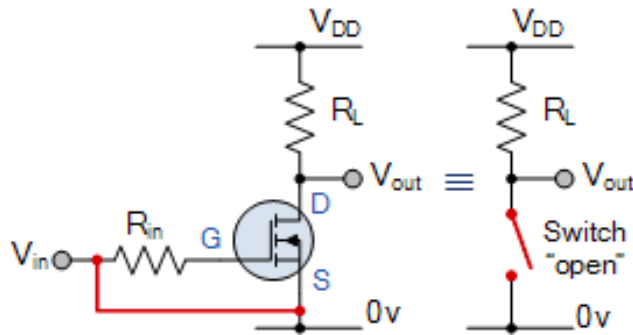
MOSFET



- MOSFET – Transistor that uses a voltage, through a capacitor, at the controlled terminal (gate) to control the current flow between two main terminals
- Threshold Voltage (V_T): Boundary between ON and OFF states
- Channel Type: N Channel (e^-) vs. P Channel (h^+); N-Channel preferred

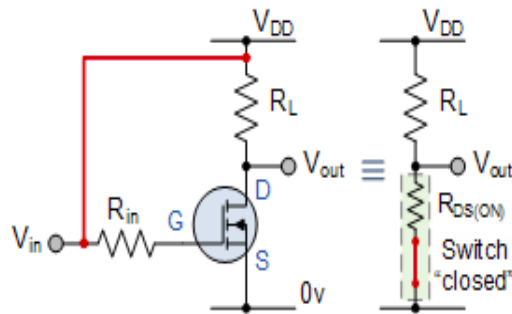
MOSFET as Switch

• “Off State”



- The input and Gate are grounded (0V)
- Gate-source voltage less than threshold voltage $V_{GS} < V_{TH}$
- MOSFET is “OFF” (Cut-off region)
- No Drain current flows ($I_D = 0$ Amps)
- $V_{OUT} = V_{DS} = V_{DD} = "1"$
- MOSFET operates as an “open switch”

• “On State”



- The input and Gate are connected to V_{DD}
- Gate-source voltage is much greater than threshold voltage $V_{GS} > V_{TH}$
- MOSFET is “ON” (linear region)
- Max Drain current flows ($I_D = V_{DD} / R_L$)
- $V_{DS} = 0V$ (ideal saturation)
- Min channel resistance $R_{DS(on)} < 0.1\Omega$
- $V_{OUT} = V_{DS} \approx 0.2V$ due to $R_{DS(on)}$
- MOSFET operates as a low resistance “closed switch”

NMOSFET Structure and Operation

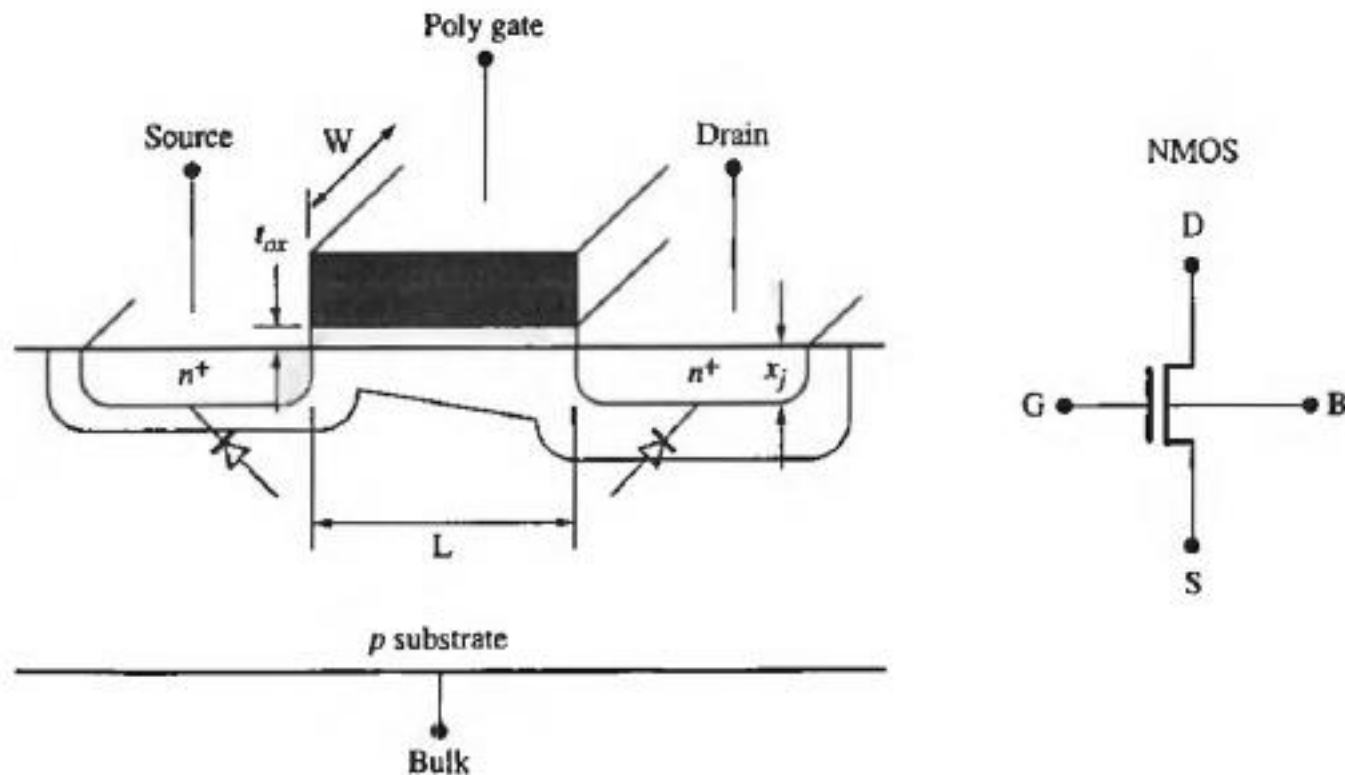


Figure 2.1

NMOS transistor structure and symbol.

PMOSFET Structure and Operation

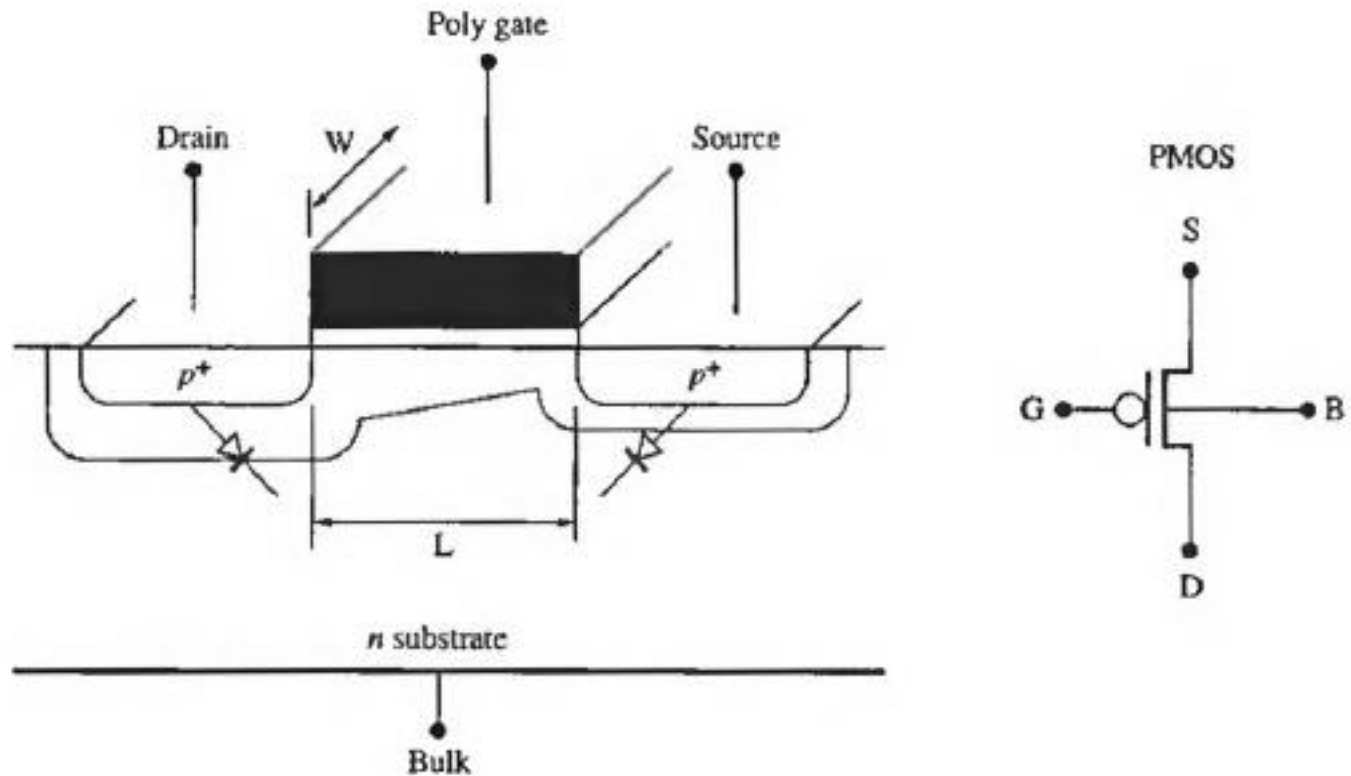
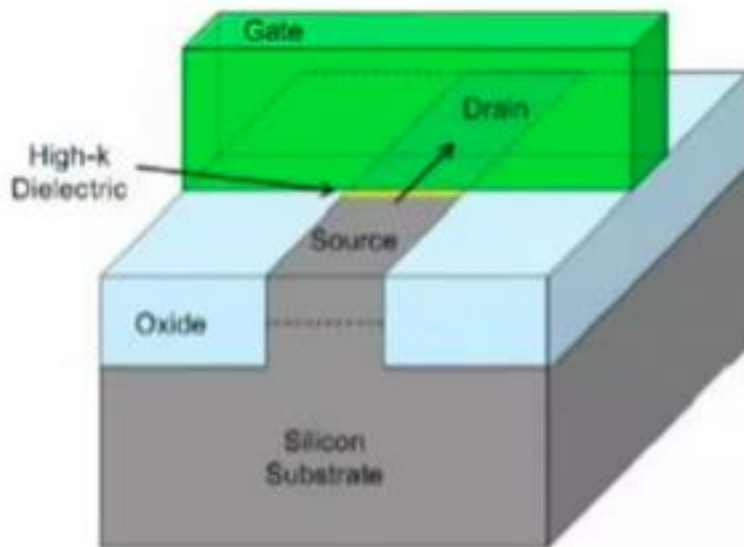


Figure 2.2
PMOS transistor structure and symbol.

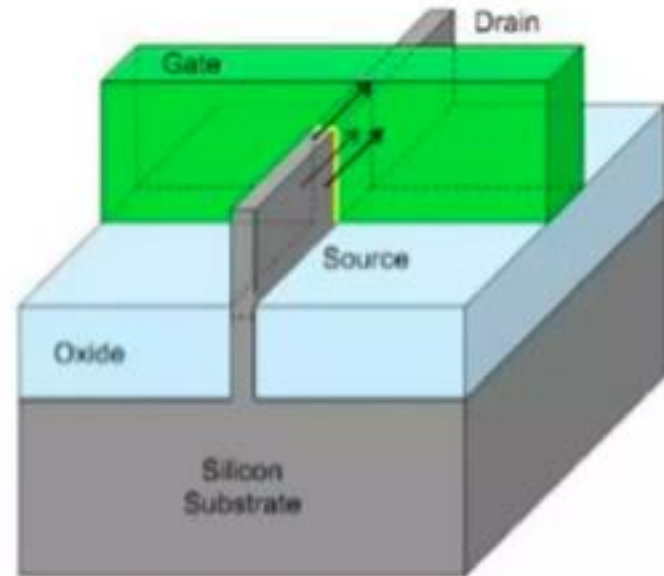
Lateral Planar vs. 3D FinFET

Traditional Planar



Traditional 2-D planar transistor form a conducting channel in the silicon region under the gate electrode when in the "on" state

3D FinFET



3-D Tri-Gate transistor form conducting channels on three sides of a vertical fin structure, providing "fully depleted" operation

CMOSFET Structure

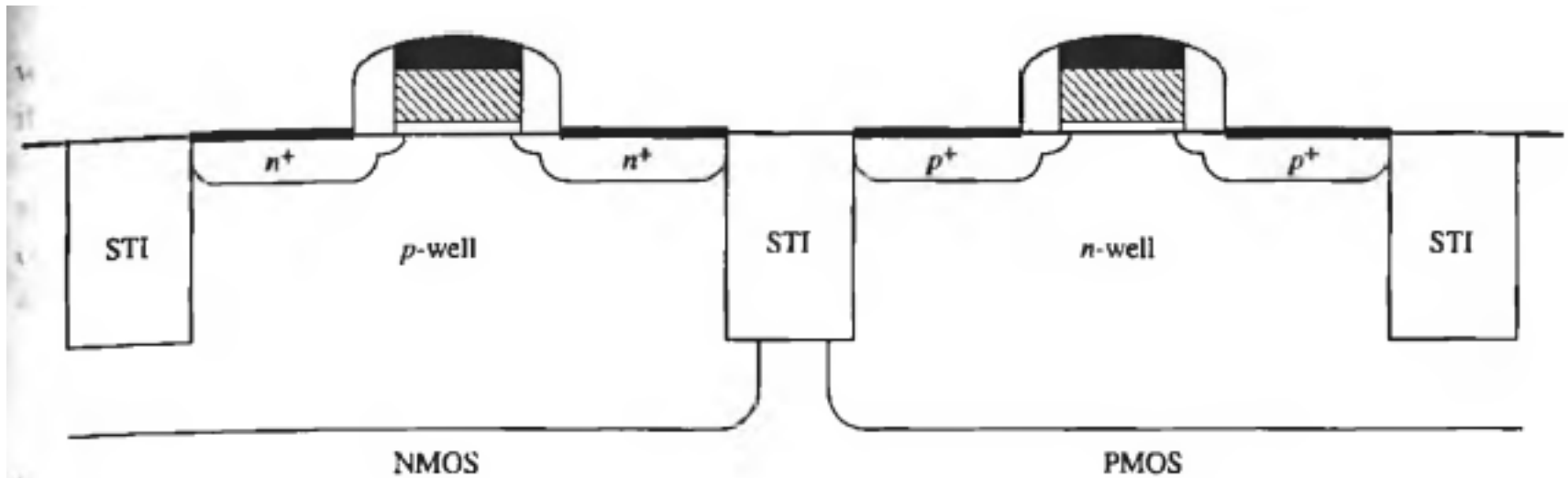
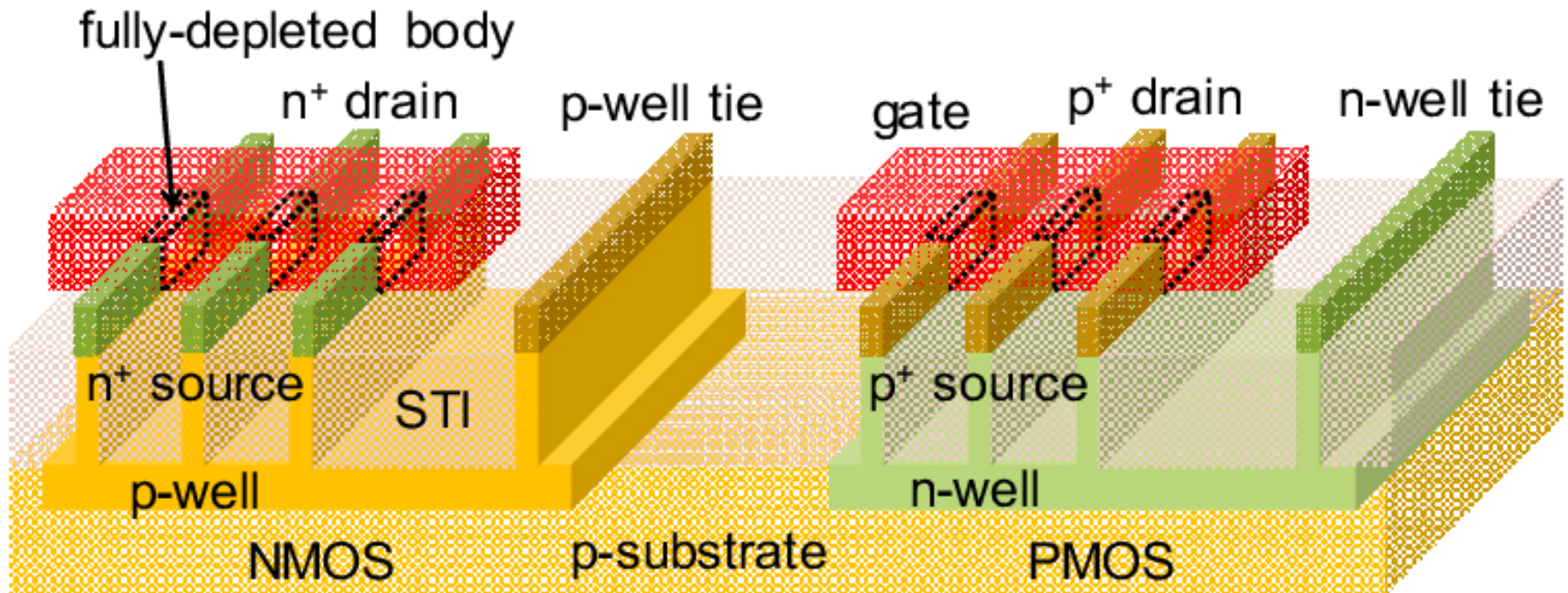


Figure 2.3
CMOS transistors structure.

CMOS FinFET Structure



Threshold Voltage

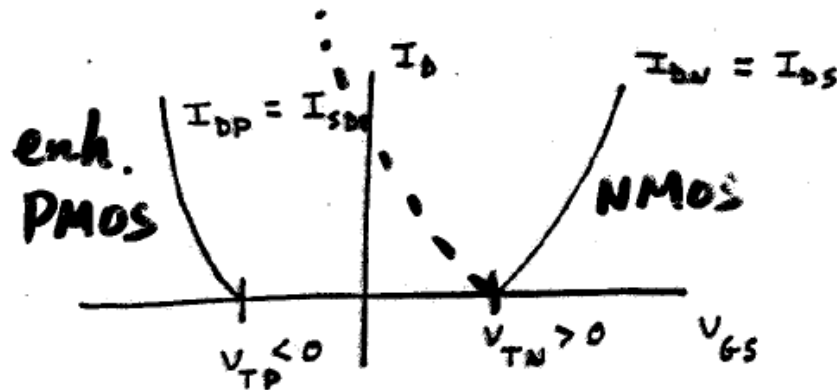


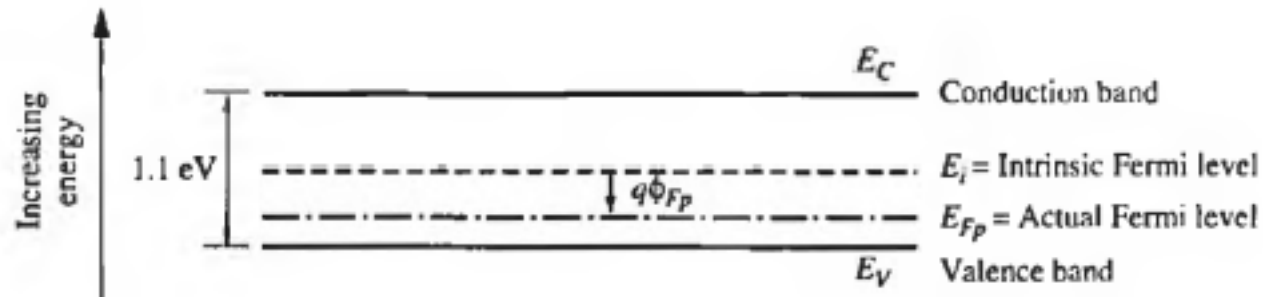
Table 2.1

Signs in threshold voltage equation

Parameter	NMOS	PMOS
Substrate	<i>p</i> -type	<i>n</i> -type
V_{T0}	+	-
ϕ_{GC} :		
n^+ polysilicon gate	-	-
p^+ polysilicon gate	+	+
Φ_F	+	-
Q_{B0}, Q_B	-	+
Q_{ox}	+	+
γ	+	-
X_d, C_{ox}	+	+
V_{SB}	+	-

Threshold Voltage

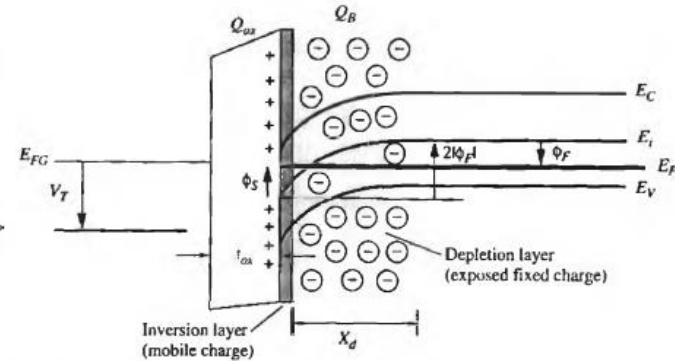
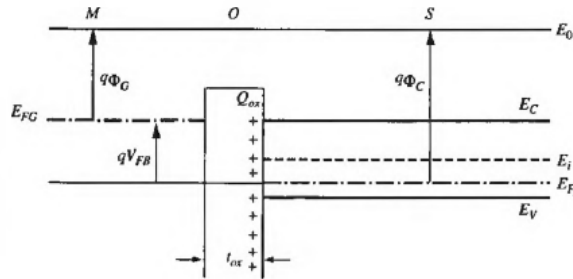
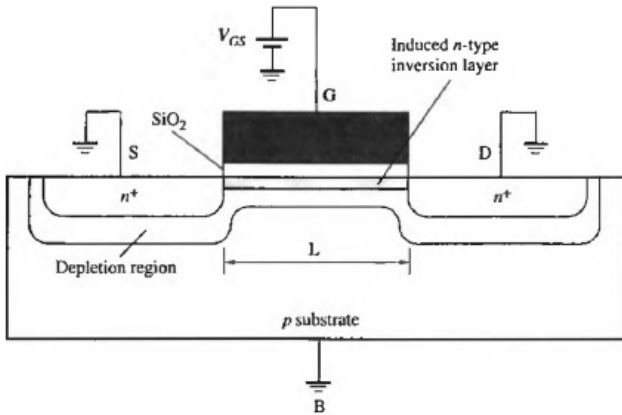
- Semiconductor Energy Band
 - Doping concentration $\longleftrightarrow$ Bulk Fermi Level



$$\phi_{Fp} = \frac{kT}{q} \ln \frac{p}{n_i} = \frac{kT}{q} \ln \frac{N_A}{n_i} > 0 \text{ (NMOS)}$$

$$\phi_{Fn} = \frac{kT}{q} \ln \frac{n_i}{n} \text{ where } n \approx N_D < 0 \text{ (PMOS)}$$

Flatband and Threshold Voltages



- Define Flatband Voltage (V_{FB}) as

$$V_{FB} = \phi_{GC} - \frac{Q_{ox}}{C_{ox}}$$
- NMOSFET: Positive gate charges are balanced by
 1. Negative ionized acceptors, N_A , in depletion region
 2. Mobile electrons in channel inversion layer

$$\phi_S = 2|\phi_{FP}| \text{ for } n_{\text{channel}} = p_{\text{bulk}} = N_A \text{ (Strong inversion)}$$

Threshold and Flatband Voltages

$$V_{T0} = \phi_{GC} + 2\phi_f - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

or

$$V_{T0} = V_{FB} + 2\phi_f - \frac{Q_{B0}}{C_{ox}}$$

- For NMOS,
 $\phi_F = \phi_{FP}$ (positive)
 $Q_{B0} < 0$ ($-Q_{B0}/C_{OX}$ is positive)
- For PMOS,
 $\phi_F = \phi_{FN}$ (negative)
 $Q_{B0} > 0$ ($-Q_{B0}/C_{OX}$ is negative)

Threshold Voltage Calculation (HJS Ex.2.1-2.4)

130nm Technology NMOSFET: Calculate V_T at $V_{SB} = 0$.

$N_A = 3 \times 10^{17} \text{ cm}^{-3}$, $t_{ox} = 2.2 \text{ nm}$, $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$ at 300K

Poly-Si gate doping $N_D = 10^{20} \text{ cm}^{-3}$,

$Q_{ox} = +2 \times 10^{10} \text{ q cm}^{-2}$

$$V_{T0} = \phi_{GC} + 2\phi_f - \frac{Q_{f0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

1. $\phi_{GC} = \phi_{G(\text{gate})} - \phi_{Fp} = -0.55 - 0.44 = -0.99 \text{ V}$

2. $2|\phi_{Fp}| = \frac{2kT}{q} \left| \ln \frac{p}{n_i} \right| = 2(0.026) \left| \ln \frac{3 \times 10^{17}}{1.45 \times 10^{10}} \right| \cong 0.88 \text{ V}$

Threshold Voltage Calculation (HJS Ex.2.1-2.4)

3. $Q_{B0} = -\sqrt{2qN_A\epsilon_{si}|-2\phi_F|}$ ← Q_{B0} has leading – for NMOS, but not for PMOS

$$Q_{B0} = -(2 \times 1.6 \times 10^{-19} \times (3)10^{17} \times 1.0 \times 10^{-12} \times |-0.88|)^{1/2}$$

$$\approx -3 \times 10^{-7} \text{ C/cm}^2$$

$$\epsilon_{ox} = 4\epsilon_0 = 3.5 \times 10^{-13} \text{ F/cm} \quad C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{(4)(8.85 \times 10^{-14})}{22 \text{ \AA}} = 1.6 \times 10^{-6} \text{ F/cm}^2$$

$$\frac{Q_{B0}}{C_{ox}} = \frac{3 \times 10^{-7}}{1.6 \times 10^{-6}} = 0.188 \text{ V}$$

4. $\frac{Q_{ox}}{C_{ox}} = \frac{2 \times 10^{10} \times 1.6 \times 10^{-19}}{1.6 \times 10^{-6}} = 0.002 \text{ V}$

$$V_{T0} = -0.99 - (-0.88) - (-0.188) - 0.002 = +0.08 \text{ V}$$

- Slightly enhancement mode ($V_T > 0$), needs a higher, more positive threshold voltage (0.4V)
- How to fix?

Threshold Voltage Calculation (HJS Ex.2.1-2.4)

Choice of N+ Poly-Si (vs. P+ Poly-Si):

$$\phi_{\text{FG(N+Poly-Si)}} = -0.55\text{V}; \phi_{\text{FG(P+Poly-Si)}} = 0.55\text{V}$$

Replacing N+ Poly-Si with P+ Poly-Si results in

$$\Delta\phi_{\text{FG}} = 1.10\text{V}$$

Hence, $V_{\text{TO}} = 1.10 + 0.08 = 1.18\text{V}$ **Too High!**

Targeted $V_{\text{TO}} = 0.4\text{V}$

Threshold Adjust Implant (HJS Ex.2-5)

Desired NFET $V'_{T0} = 0.4V$. What is N_I needed?

Implant acceptor ions to increase NFET V_{T0}

$$Q_I = q N_I = C_{ox} (V'_{T0} - V_{T0}) = C_{ox} \Delta V_{T0}$$

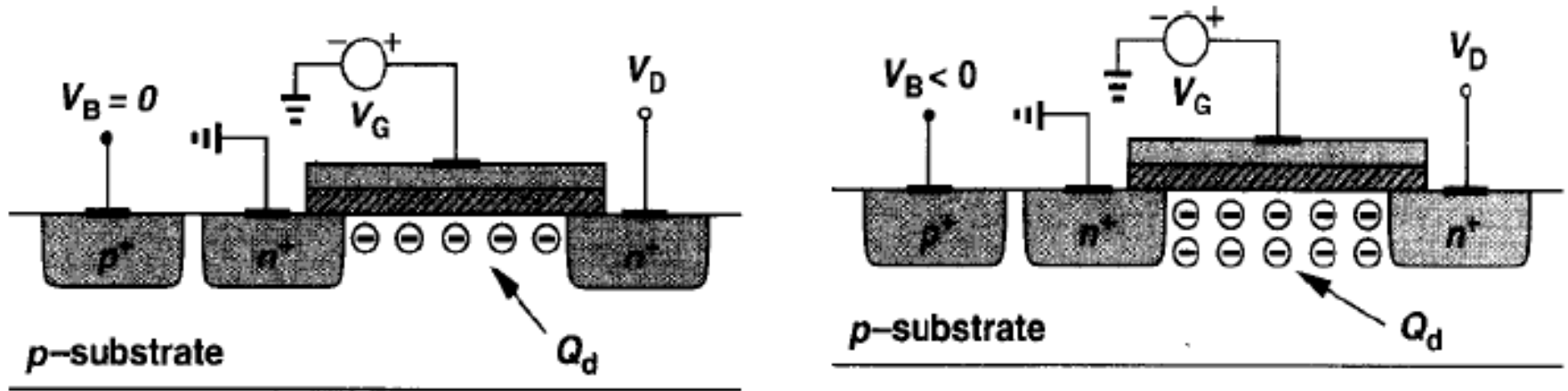
We know that V_T will be shifted by $Q_I / C_{ox} = qN_I / C_{ox}$

$$\therefore N_I = \frac{Q_I}{q} = \frac{C_{ox} \Delta V}{q} = \frac{C_{ox} (0.4 - (+0.08))}{1.6 \times 10^{-19}} = 3.2 \times 10^{12} \frac{\text{ions}}{\text{cm}^2}$$

Shallow ($0.1\mu\text{m}$ deep) box implant profile:

$$\begin{aligned} N_A = N_I / x_j &= (3.2 \times 10^{12} \text{cm}^{-2}) / (0.1 \times 10^{-4} \text{cm}) \\ &= 3.2 \times 10^{17} \text{cm}^{-3} \end{aligned}$$

Body Bias Effect



- Reverse biasing source/substrate junction increases space charges under channel in the substrate, hence increasing threshold voltage

$$V_{T0} + \gamma(\sqrt{V_{SB} + |2\phi_F|} - \sqrt{|2\phi_F|})$$

where $\gamma = \frac{1}{C_{ox}} \sqrt{2q\epsilon_{si} N_A}$ body-effect coefficient

Body Bias Effect (HJS Ex. 2.3+)

- Since $\gamma = \sqrt{2q\epsilon_s N_A / C_{ox}}$

$$\gamma = \sqrt{(2)(1.6 \times 10^{-19})(11.7)(8.85 \times 10^{-14})(3 \times 10^{17}) / 1.6 \times 10^{-6}}$$

$$\approx 0.2 \text{ V}^{1/2}$$

$$V_T = \phi_{GC} + 2\phi_F - \frac{Q_B}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

$$= \phi_{GC} + 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} - \frac{Q_B - Q_{B0}}{C_{ox}}$$

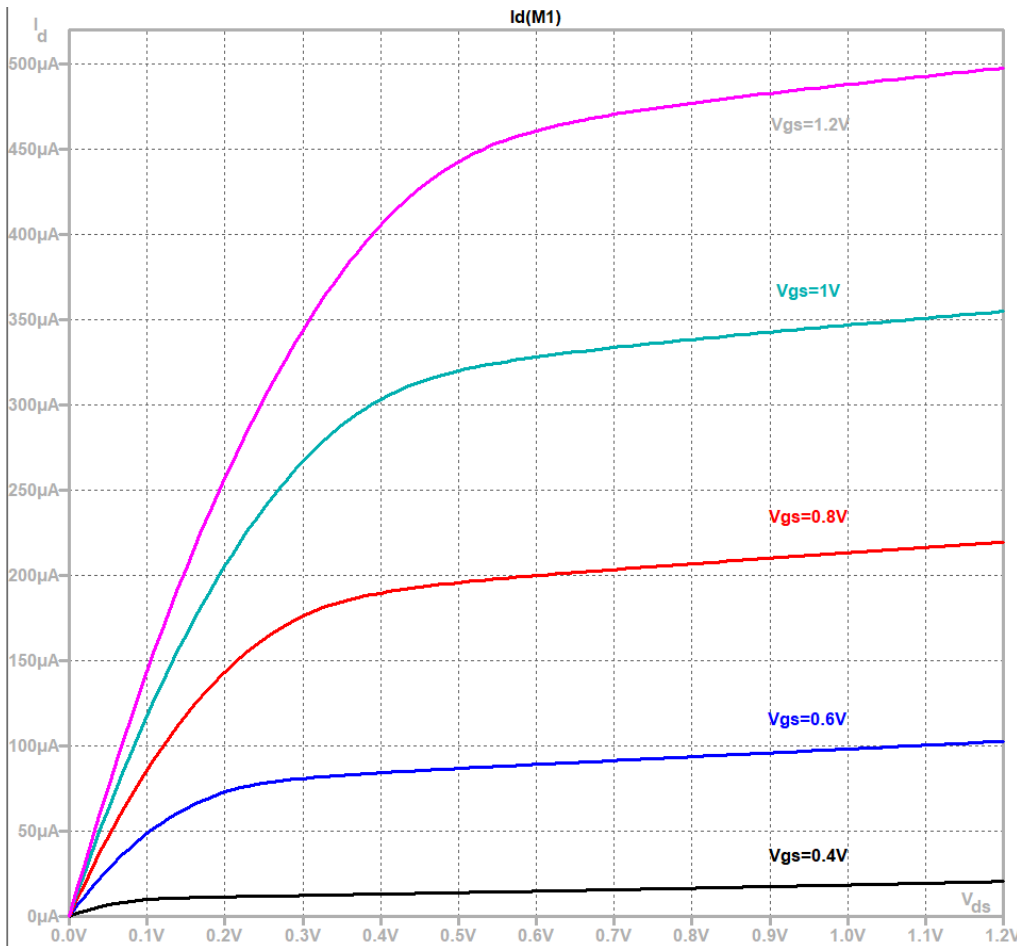
$$= V_{T0} + \gamma(\sqrt{V_{SB} + |2\phi_F|} - \sqrt{|2\phi_F|})$$

When $V_{SB} = V_{DD} = 1.2\text{V}$,

$$V_T = 0.08 + 0.2 [(1.2+0.88)^{1/2} - (0.88)^{1/2}] = 0.18\text{V}$$

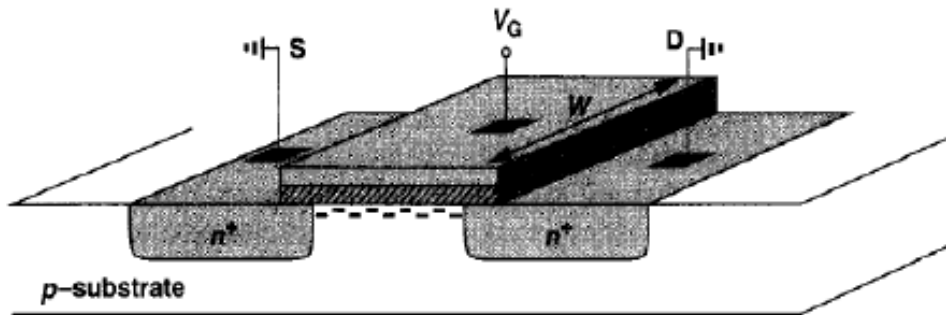
Or $V_T = 0.5\text{V}$ with threshold adjust implant

130nm DSM CMOS Tech

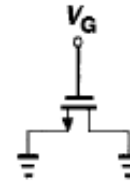


130nm CMOS
 $W=1.0\mu\text{m}, L=0.13\mu\text{m}$ NFET
 $V_{TH} = 0.33\text{V}$
 $I_{DSsat} = 475\mu\text{A}$ ($V_{GS}=1.2\text{V}$)
 $K' = 423\mu\text{A}/\text{V}^2$
 $C_G = 2.0\text{fF}$
 $R_{ON} = 1.5\text{k}\Omega$ ($V_{GS}=1.2\text{V}$)

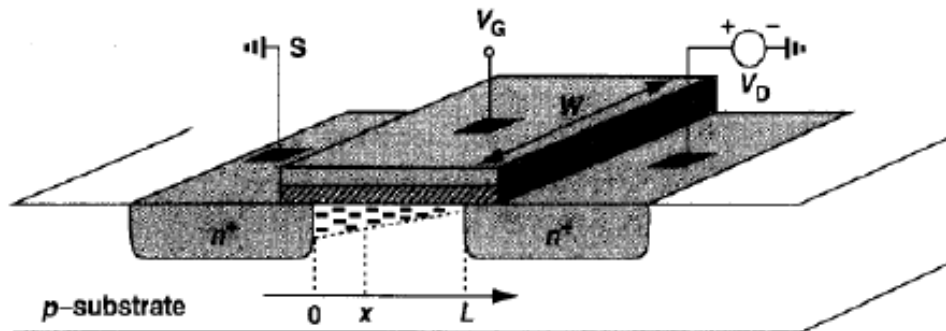
MOSFET Current



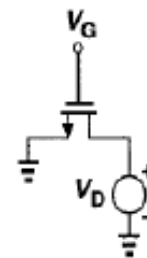
(a)



$V_{GS} > V_{th}$ channel created,
no current yet!



(b)



$V_{DS} > 0$ carriers start flowing
From source into the drain

$$Q_n(y) = C_{ox} [V_{GS} - V(y) - V_T]$$

Charge per unit area

MOSFET 1st Order I-V Characteristics

How do we determine IV characteristics ?

Basic Equations: $Q = C V$

$$V = I R$$

At a distance y along the channel,

$$Q_I(y) = C_{ox} [V_{GS} - V(y) - V_T]$$

$$[Q] = \text{Coulombs/cm}^2$$

$$[C_{ox}] = \text{Farads/cm}^2$$

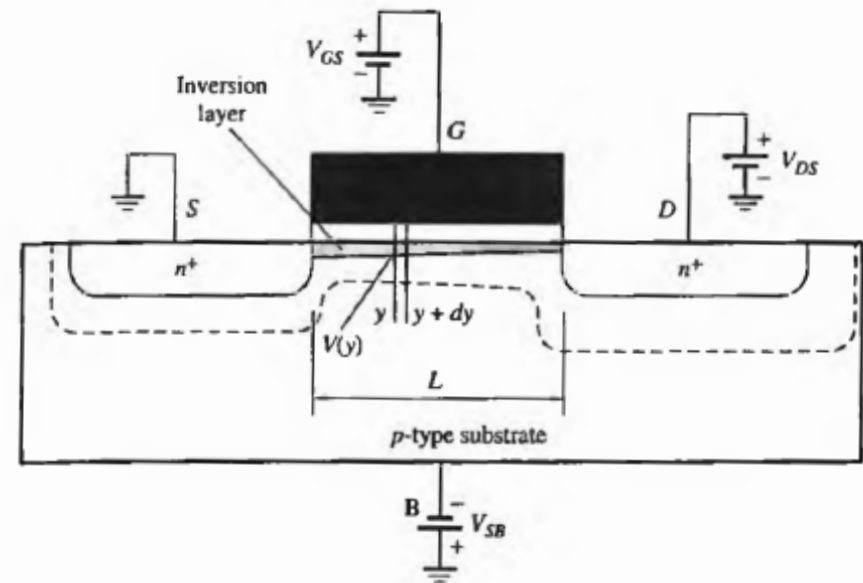
$$dV = I_D dR$$

$$dR = R_S \frac{dy}{W} = \frac{dy}{\mu_n Q_I(y) W}$$

$$\text{so } I_D \int_0^L dy = W \mu_n C_{ox} \int_0^{V_{DS}} [V_{GS} - V(y) - V_T] dV$$

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$I_D \text{ const}$



$$I_D = f(V_{GS}, V_{DS})$$

MOSFET 1st Order I-V Characteristics

$$I_D = \frac{k}{2} [2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

k = device conductance parameter

$$= k' (W/L)$$

k' = process conductance parameter

$$= \mu_n C_{ox}$$

OK for $V_{GS} \geq V_T$, $V_{DS} \leq V_{GS} - V_T$

"linear" or resistive region, low V_{DS}

For $V_{DS} \geq V_{GS} - V_T$, pinched off channel

$$I_D \approx \frac{k}{2} (V_{GS} - V_T)^2 \quad \text{saturation region}$$

$$I_D = \frac{k}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

λ = channel length modulation parameter

$$\approx 0.1 \text{ to } 0.01 \text{ V}^{-1}$$

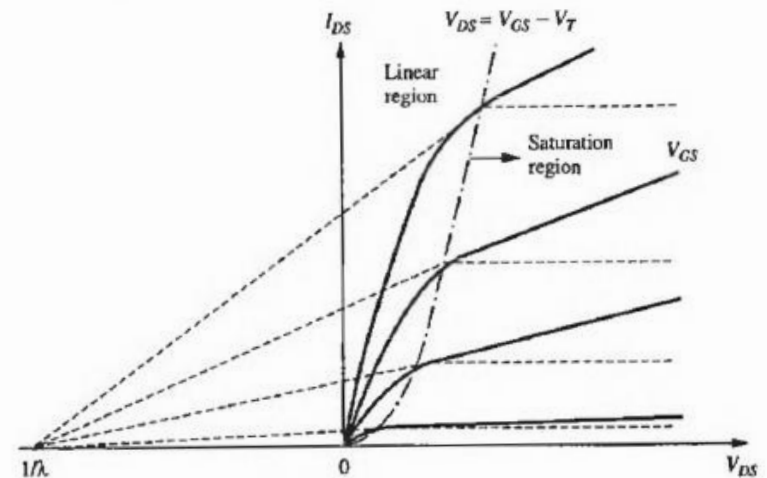


Figure 2.10
NMOS device $I_D - V_{DS}$ characteristics.

MOSFET I-V Regions

Three regions of operation:

1. Cutoff $V_{GS} < V_T$

$$I_D \approx 0$$

2. Linear or resistive $V_{GS} \geq V_T$

$$I_D = \frac{k}{2} [2 (V_{GS} - V_T) V_{DS} - V_{DS}^2]$$

$$V_{DS} \leq V_{GS} - V_T$$

triode

3. Saturation $V_{GS} \geq V_T$

$$I_D = \frac{k}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

$$V_{DS} \geq V_{GS} - V_T$$

pentode

$$2 I_D \approx k (V_{GS} - V_T)^2$$

MOSFET I-V Regions

Resistive region:

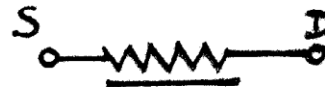
For $V_{DS} \approx 0$, $I_D \approx k(V_{GS} - V_T)V_{DS}$

G_C = channel conductance = $\frac{dI_D}{dV_{DS}}$

$\approx k(V_{GS} - V_T)$

R_C = channel resistance

$$\approx \frac{1}{k(V_{GS} - V_T)}$$



$$\sqrt{2I_D} \approx k(V_{GS} - V_T)$$

$$I_D \propto V_{DS}$$

$$\propto (V_{GS} - V_T)$$

Saturation region:

$I_D \approx \frac{k}{2}(V_{GS} - V_T)^2$ independent of V_{DS} ($\lambda = 0$)

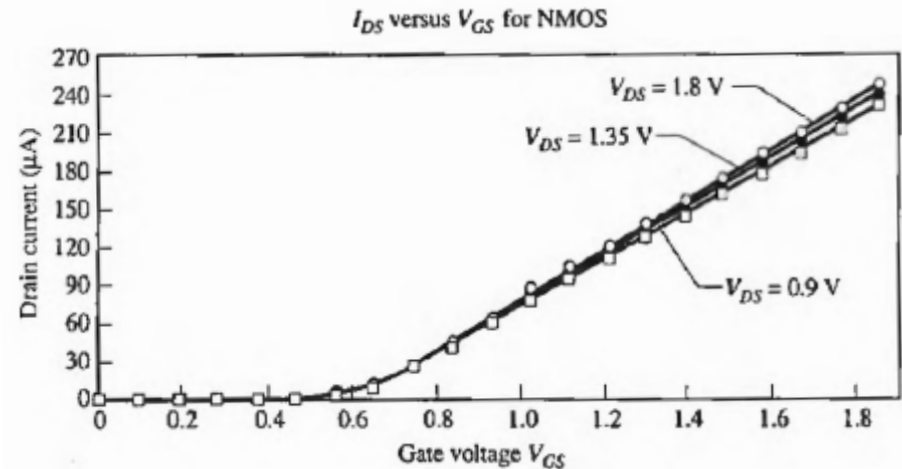
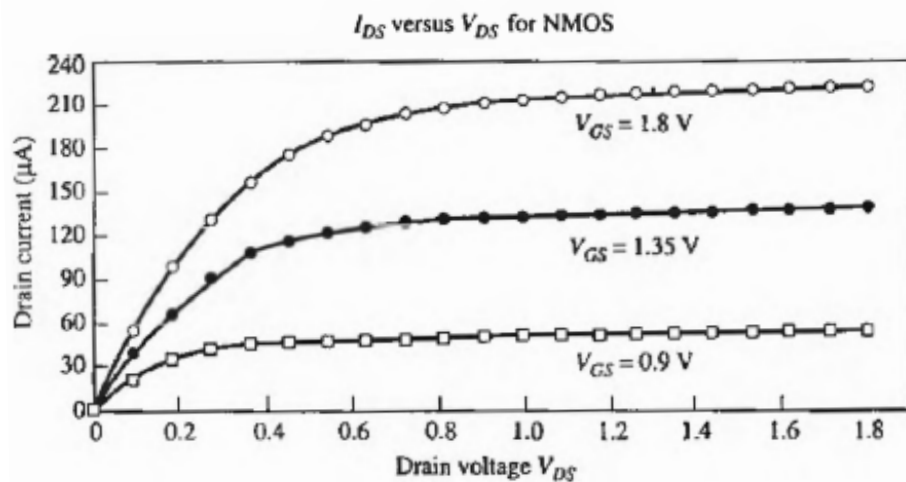


$$I_D \propto (V_{GS} - V_T)^2$$

Cutoff region:



0.18 μm NMOSFET I-V Characteristics



Deviations from long-channel theory:

- $I_{DS} \propto (V_{GS} - V_T)$ instead of $(V_{GS} - V_T)^2$ in saturation

Electric Fields in Short-Channel MOSFET

1980	1995	2001
$E_y = \frac{5V}{5\mu m} = 10^4 V/cm$	$E_y = \frac{3.3V}{0.35\mu m} = 9.4 \times 10^4 V/cm$	$E_y = \frac{1.2V}{0.1\mu m} = 1.2 \times 10^5 V/cm$

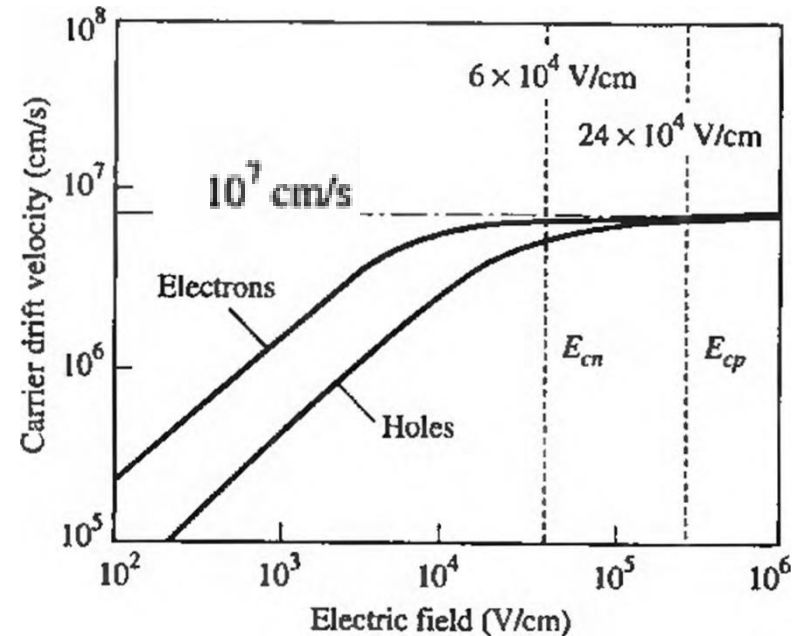
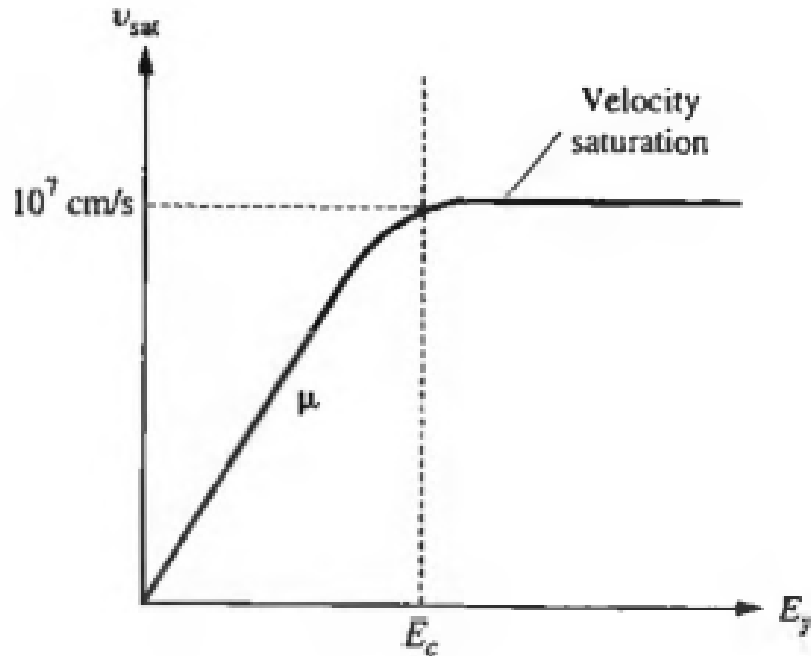
Horizontal Electric field (between the source and drain) = $V_{DS} (=V_{dd}) / L$

1980	1995	2001
$E_x = \frac{5V}{1000 \text{ \AA}} = 50 \times 10^4 V/cm$	$E_x = \frac{3.3V}{75 \text{ \AA}} = 4.4 \times 10^6 V/cm$	$E_x = \frac{1.2V}{22 \text{ \AA}} = 5.5 \times 10^6 V/cm$

Vertical Electric field (between the gate and channel) = $V_{GS} (=V_{dd}) / t_{ox}$

- Large electric field causes early carrier velocity saturation in channel of short-channel MOSFETs

Carrier Velocity Saturation



- Carriers reach limiting or saturation velocity (10^7 cm/s in Si) at high electric field
- Electrons saturate at a lower electric field than holes

$$v = \mu_e \frac{E_y}{\left(1 + \frac{E_y}{E_c}\right)} \quad E_y < E_c$$

$$v = v_{sat} \quad E_y \geq E_c$$

$E_{cn} = 6 \times 10^4 \frac{\text{V}}{\text{cm}}$ for electrons
 $E_{cp} = 24 \times 10^4 \frac{\text{V}}{\text{cm}}$ for holes

MOSFET Saturation Current

Qualitative I-V Curves in Velocity Saturation Conditions

